Week 9 - Sample Solutions for Selected Questions from Textbook for Self-Practices

Notes to students:

- These solutions are prepared to assist your self-practices and self-checking. There is absolutely no intention to relate to any assessment or exam.
- If you see any error in the solution, please send an email to me (cheeshen.lim@xjtlu.edu.cn) so that I can check and update the solution.
- Reminder: please always show full calculation/derivation steps in exam.

Week9-Self Practice Q1 (9th ed., Q35)

Given that $\mathbf{I}_{10} = 4\angle 35°$ A ; $\mathbf{V} = 10\angle 35°$ V; $\mathbf{I} = 2\angle 35°$ A

(a) $\mathbf{V}$ is in-phase with its current $\mathbf{I}_{10}$, so it can be deduced that the circuit element is a resistor, of value

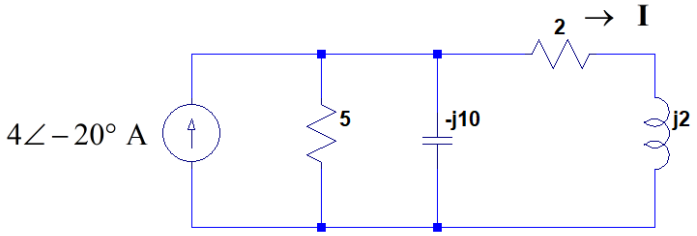
$$R = \frac{\mathbf{V}}{\mathbf{I}_{10}} = \frac{10\angle 35°}{4\angle 35°} = \frac{10}{4} = 2.5\ \Omega$$

(b) We know that $\mathbf{V}_s = 10\mathbf{I}_{10\Omega} + \mathbf{V} + \mathbf{V}_{25\Omega}$

$$\mathbf{I}_{25} = \mathbf{I}_{10} - \mathbf{I} = 4\angle 35° - 2\angle 35° = 2\angle 35°\ \text{A}$$
$$\mathbf{V}_{25} = 25(2\angle 35°) = 50\angle 35°\ \text{V}$$
$$\mathbf{V}_s = 10\mathbf{I}_{10} + \mathbf{V} + \mathbf{V}_{25} = 40\angle 35° + 10\angle 35° + 50\angle 35°$$
$$= 100\angle 35°\ \text{V}$$

Week9-Self Practice Q2 (9th ed., Q44)

Draw the circuit,



Current $\mathbf{I}$ flows through the series combination $2 + j2\ \Omega$. It can be calculated through "current division" technique,

$$\mathbf{I} = \left(4\angle -20^o\right)\frac{5 \,||\, -j10}{5 \,||\, -j10 + (2 + j2)}$$

$$= \left(4\angle -20^o\right)\frac{\dfrac{5(-j10)}{5 - j10}}{\dfrac{5(-j10)}{5 - j10} + (2 + j2)}$$

$$= \left(4\angle -20^o\right)\frac{-j50}{-j50 + (2 + j2)(5 - j10)}$$

$$= \left(4\angle -20^o\right)\frac{-j50}{30 - j60}$$

$$= \left(4\angle -20^o\right)\frac{50\angle -90^o}{67.08\angle -63.43^o}$$

$$\mathbf{I} = \frac{4 \times 50}{67.08}\angle(-20^o - 90^o + 63.43^o)$$

$$= 2.98\angle -46.57^o \text{ A}$$

$$i(t) = 2.98\cos(100t - 46.57^o) \text{ A}$$

Week9-Self Practice Q3 (9th ed., Q50)

Question never asks for any specific method. Hence we can choose one that can easily relate to the answers we want ($\mathbf{I}_1$, $\mathbf{I}_1$) → mesh analysis seems to be most appropriate (but other methods can also be used).

$$-10\angle -80^o + j30\mathbf{I}_1 + 55(\mathbf{I}_1 - \mathbf{I}_2) + 4\angle 0^o = 0$$
$$-4\angle 0^o + 55(\mathbf{I}_2 - \mathbf{I}_1) - j20\mathbf{I}_2 + 2\angle -23^o = 0$$

Grouping the terms,

$$(55 + j30)\mathbf{I}_1 - 55\mathbf{I}_2 = 10\angle -80^o - 4\angle 0^o$$
$$-55\mathbf{I}_1 + (55 - j20)\mathbf{I}_2 = 4 - 2\angle -23^o$$

Solving,

$$\mathbf{I}_1 = 0.610\angle -156.6^o \text{ A}$$
$$\mathbf{I}_2 = 0.534\angle -136.4^o \text{ A}$$

Week9-Self Practice Q4 (9th ed., Q62)

Question never asks for any specific method, but since the answers are voltage phasors, it seems appropriate and most direct to use Nodal analysis.

$$33\angle 3° - 51\angle -91° = \frac{\mathbf{V}_1}{j3} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{-j5}$$

$$51\angle -91° = \frac{\mathbf{V}_2}{2} + \frac{\mathbf{V}_2 - \mathbf{V}_1}{-j5}$$

$$33\angle 3^o - 51\angle -91^o = \frac{\mathbf{V}_1}{j3} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{-j5}$$

$$51\angle -91^o = \frac{\mathbf{V}_2}{2} + \frac{\mathbf{V}_2 - \mathbf{V}_1}{-j5}$$

$$(j3)(j5)\left(33\angle 3^o - 51\angle -91^o\right) = j2\mathbf{V}_1 + j3\mathbf{V}_2$$

$$(2)(j5)\left(51\angle -91^o\right) = 2\mathbf{V}_1 + (-2 + j5)\mathbf{V}_2$$

$$-15\left(33\angle 3^o - 51\angle -91^o\right) = j2\mathbf{V}_1 + j3\mathbf{V}_2$$

$$-10\left(51\angle -91^o\right) = j2\mathbf{V}_1 + (-5 - j2)\mathbf{V}_2$$

$$-15\left(33\angle 3^o - 51\angle -91^o\right) + 10\left(51\angle -91^o\right) = (5 + j5)\mathbf{V}_2$$

$$\mathbf{V}_2 = \frac{-15\left(33\angle 3^o - 51\angle -91^o\right) + 10\left(51\angle -91^o\right)}{(5 + j5)}$$

$$\mathbf{V}_2 = 198\angle -156.7^o \text{ mV}$$

Solve the above and we can get the answer (take note on the units; the given currents are in mA)

$$-10\left(51\angle -91^o\right) = j2\mathbf{V}_1 + (-5 - j2)\left(198\angle -156.7^o\right)$$

$$\mathbf{V}_1 = \frac{(5 + j2)\left(198\angle -156.7^o\right) - 10\left(51\angle -91^o\right)}{j2}$$

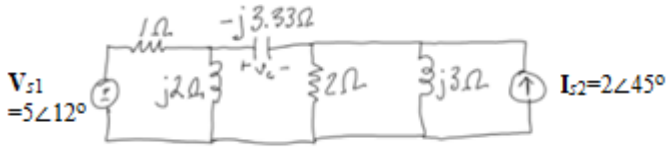
$$= 391\angle 108.3^o \text{ mV}$$

Week9-Self Practice Q5 (9th ed., Q76)

(a)

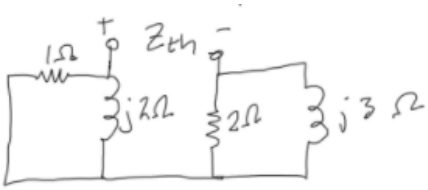
$$X_{L1} = j\omega L_1 = j(20)(0.1) = j2\,\Omega$$
$$X_{L2} = j\omega L_2 = j(20)(0.15) = j3\,\Omega$$
$$X_C = \frac{1}{j\omega C} = \frac{1}{j(20)(0.015)} = -j3.33\,\Omega$$

$$\mathbf{V}_{s1} = 5\angle 12^o \text{ V}$$
$$\mathbf{I}_{s2} = 2\angle 45^o \text{ A}$$



b) To find the Thevenin equivalent, first calculate $\mathbf{Z}_{Th}$

"short the independent voltage source and open the independent current source, remove the load (capacitor) connected to the load/output terminal, determine the equivalent impedance"



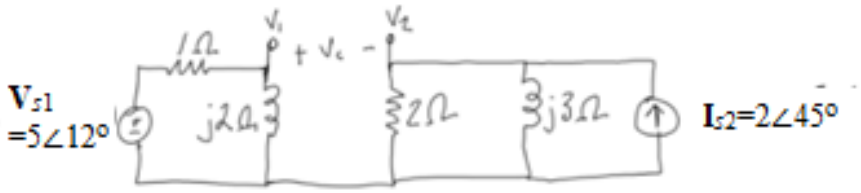
$$\mathbf{Z}_{Th} = \frac{(1)(j2)}{1+j2} + \frac{(2)(j3)}{2+j3} = 2.18 + j1.32 = 2.55\angle 31.2^o\,\Omega$$

Then, calculate the $\mathbf{V}_{Th} = \mathbf{V}_{oc}$

"Insert back the independent voltage source and the independent current source; the load (capacitor) remains disconnected from the load terminal. Determine the voltage phasor across the load/output terminal"

Calculate the Thevenin equivalent voltage across the capacitor by open-circuiting the capacitor.

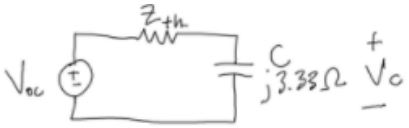
$$\mathbf{Z}_{Th} = \frac{(1)(j2)}{1+j2} + \frac{(2)(j3)}{2+j3} = 2.18 + j1.32 = 2.55\angle 31.2^o\,\Omega$$



$$\mathbf{V}_1 = \frac{j2}{1+j2}\mathbf{V}_{s1} = \frac{j2}{1+j2}\left(5\angle 12^\circ\right) = 4.47\angle 38.57^\circ \text{ V}$$

$$\mathbf{V}_2 = \mathbf{I}_{s2}\left(2 \parallel j3\right) = \left(2\angle 45^o\right)\left(\frac{(2)(j3)}{2+j3}\right) = 3.33\angle 78.69^\circ \text{ V}$$

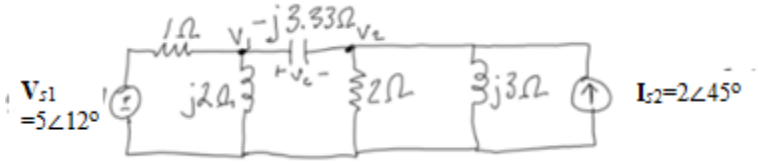
$$\begin{aligned}\mathbf{V}_{Th} &= \mathbf{V}_{oc} \\ &= \mathbf{V}_1 - \mathbf{V}_2 \\ &= 4.47\angle 38.57^\circ - 3.33\angle 78.69^\circ \\ &= 3.497 + j2.787 - (0.653 + j3.264) \\ &= 2.885\angle -9.49^\circ\end{aligned}$$



$$\boldsymbol{V}_C = \left(\frac{-j3.33}{2.18 + j1.32 - j3.33}\right)(2.885\angle -9.49^o)$$
$$= 3.23\angle -56.88^o \text{ V}$$

$$v_C(t) = 3.23\sin(20t - 56.88^\circ) \text{ V}$$

c)



$$\frac{\mathbf{V}_1 - \mathbf{V}_{s1}}{1} + \frac{\mathbf{V}_1}{j2} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{-j3.33} = 0$$

$$\frac{\mathbf{V}_1 - 5\angle 12^o}{1} + \frac{\mathbf{V}_1}{j2} + \frac{3.23\angle -56.88^o}{-j3.33} = 0$$

$$(1 - j0.5)\mathbf{V}_1 = 5\angle 12^o + \frac{3.23\angle -56.88^o}{j3.33}$$

$$(1 - j0.5)\mathbf{V}_1 = 5\angle 12^o + \frac{3.23}{3.33}\angle -146.88^o$$

$$(1 - j0.5)\mathbf{V}_1 = 4.078 + j0.510$$

$$\mathbf{V}_1 = \frac{4.11\angle 7.12^o}{1.118\angle -26.57^o}$$

$$= 3.68\angle 33.69^o \text{ V}$$

$$\mathbf{I}_{s1} = \frac{\mathbf{V}_{s1} - \mathbf{V}_1}{1}$$

$$= 5\angle 12^o \text{-} 3.68\angle 33.69^o$$

$$= 4.891 + j1.040 - (3.062 + j2.041)$$

$$= 2.09\angle -28.69^o \text{ A}$$

$$i_{s1}(t) = 2.09\sin(20t - 28.69^o) \text{ A}$$